

Lab 1 – RedFlag Descriptive Paper

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1 Introduction

The rapid growth of digital media has significantly changed how people consume information, but it has also introduced a serious societal issue: the widespread dissemination of misinformation. With the rise of social media platforms, online news outlets, and user-generated content, individuals are exposed to vast amounts of information daily, much of which may be misleading, biased, or entirely false. According to the Pew Research Center (2024), a large percentage of Americans regularly encounter misleading or inaccurate information online. This issue has far-reaching consequences, including eroding public trust, spreading harmful narratives, and driving poor decisions in areas such as health, politics, and finance.

Current solutions to misinformation often rely on manual fact-checking or user initiative, requiring individuals to verify content themselves. These approaches are inefficient, time-consuming, and often unrealistic given how quickly people consume information. Additionally, many users lack the tools or expertise to critically evaluate source credibility in real time. Therefore, a system is needed to automatically help users identify potentially misleading content as they browse the internet without disrupting their experience.

To address this issue, this project introduces RedFlag, a browser extension and desktop application that automatically detects and flags potentially misleading or inaccurate content in real time. RedFlag runs in the background while users browse, analyzes content, and provides contextual warnings when needed. Rather than requiring active user input, the system integrates seamlessly into the browsing experience, offering clear explanations and optional interaction with an AI-powered assistant. By combining automation, user control, and transparency, RedFlag aims to empower users to make more informed decisions and reduce the impact of misinformation in digital spaces. The product will be demonstrated through a prototype that

highlights its core functionality, usability, and effectiveness in addressing this growing societal problem.

2 Product Description

RedFlag is a cross-platform misinformation detection and bias-awareness tool that helps users evaluate online claims without leaving their normal browsing experience. The product operates in the background while a user views web pages or social media content. When RedFlag identifies a claim that may be false, misleading, outdated, or missing important context, it displays a subtle warning near the questionable material. The user can open that warning to review a credibility score, a short explanation, and links to evidence from reliable sources.

RedFlag's goal is not to decide what a user must believe. Instead, it reduces the time and effort required to investigate a questionable claim before reacting to or sharing it. RedFlag combines automated content analysis with external evidence and source information, then presents the results in a clear, nonintrusive interface. For politically divisive material, its BiasStar feature also presents relevant viewpoints from across the political spectrum without assigning ideological labels to the user. Together, these capabilities encourage users to pause, review the available evidence, and make a better-informed decision.

2.1 Key Product Features and Capabilities

RedFlag monitors supported web content as it appears in the browser and identifies factual claims that may require additional review. Automatic detection removes the need to manually copy a claim into a search engine or visit several fact-checking websites. The analysis runs in the background, letting the user keep browsing while RedFlag evaluates the selected text. Questionable statements are marked with a small, visible indicator rather than a full-screen interruption. The warning is designed to attract attention before the user accepts or shares the

claim while still allowing normal interaction with the page. Selecting the warning opens the detailed RedFlag result.

Each analyzed claim receives an easy-to-understand misinformation or credibility score based on the available evidence, source characteristics, and agreement with reputable reporting or fact-checking information. RedFlag also provides direct links to the supporting sources so the user can verify the explanation instead of relying only on an automated label. BiasStar identifies politically divisive content and provides multiple relevant perspectives. It does not label the user or declare that one ideology is correct. Its purpose is to expose users to evidence and interpretations that may not appear in their usual information feed, helping reduce selective exposure and online echo chambers.

Users can request a short explanation of why content was flagged. The explanation identifies the claim under review, summarizes conflicting or supporting evidence, and notes issues such as missing context, outdated information, or weak sourcing. This gives the user enough information to understand the warning while keeping the initial alert brief. The user can dismiss a warning, open its sources, or provide feedback when a result appears inaccurate or unclear. These controls keep RedFlag from interrupting the browsing experience and provide information that can help the development team evaluate usability and improve future analysis.

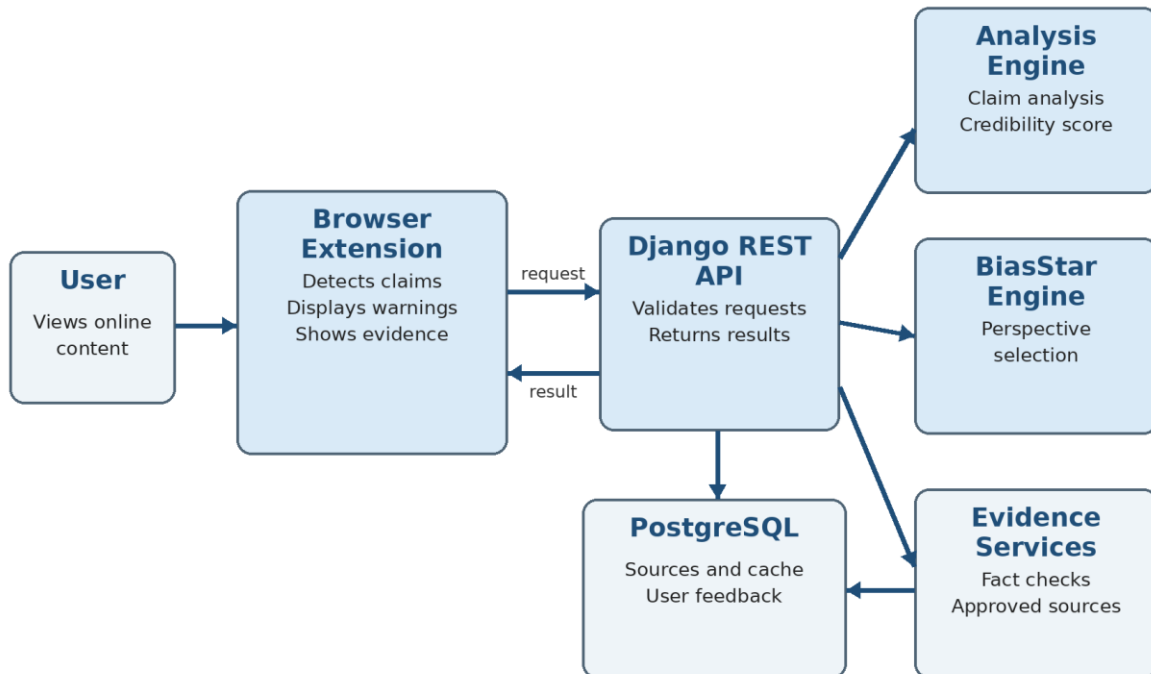
2.2 Major Components (Hardware/Software)

RedFlag follows a client-server structure based on the project's Major Functional Component Diagram (MFCD), shown in Figure 1. The client component detects content and displays results in the user's browser. The server components process claims, perform credibility and bias analysis, retrieve outside evidence, and return a response. A data store supports source records, cached results, system configuration, and user feedback. This separation keeps the

browser component lightweight while letting the analysis services be updated without major changes to the user interface.

Figure 1

RedFlag Major Functional Component Diagram



The user-side hardware is an internet-connected desktop or laptop computer that can run a supported web browser and the RedFlag extension. Normal use requires no specialized hardware. During prototype development and testing, team computers will run the browser extension and prepared test pages. The real-world product would use an internet-accessible application server or cloud-hosted environment to receive analysis requests, communicate with external information services, and store the limited data required for operation. Server capacity could be increased as the number of users and analyzed claims grows.

The browser extension is the user-facing component. It extracts supported visible text, submits claims for analysis, places RedFlag indicators on the page, and displays scores,

explanations, source links, and BiasStar results. When relevant text appears inside an image, an optical character recognition component may convert it into text before analysis. The prototype extension will be developed with JavaScript, HTML, CSS, and the Chrome Extension APIs.

A secure application programming interface receives requests from the extension and returns structured results. It coordinates the client, analysis engine, evidence services, and data store, validates requests, and limits unnecessary collection of browsing information. The back end will be written in Python using Django and Django REST Framework, and PostgreSQL will store source records, cached results, application settings, and user feedback.

Natural language processing separates factual claims from opinion or ordinary page text. The credibility component compares the claim with available evidence and produces the score and explanation. The BiasStar component determines whether multiple political perspectives would help the user understand the topic. Fact-checking databases, reputable news sources, and other approved information services provide evidence for evaluating claims. RedFlag records the source links associated with a result so users can examine the original material. The design minimizes stored browsing information and keeps only the data needed to produce, evaluate, or improve a result.

3 Identification of Case Study

RedFlag is being developed for people who regularly encounter news, commentary, health claims, political content, and other factual statements online but do not have the time or expertise to investigate every claim independently. The product is intended to reduce the effort required to check questionable information while preserving the user's ability to review the evidence and decide what to believe. College students are an especially appropriate initial audience because they frequently use social media and online sources for both academic and

personal information, and they can provide detailed feedback about whether the warnings are clear, useful, and nonintrusive.

The initial case study group will consist of a small group of Old Dominion University undergraduate students who use a supported browser and regularly view online news or social-media content. Participants will complete representative browsing tasks with the RedFlag prototype, open selected warnings, review the displayed scores and sources, and provide feedback through a short questionnaire. Their feedback will focus on the clarity of the warning indicator, the usefulness of the explanation and evidence, the perceived accuracy of results, and whether the extension interrupts normal browsing.

The following hypothetical day-in-the-life scenario illustrates how one participant would use the prototype during the case study. Maya, an Old Dominion University undergraduate, opens Google Chrome on her laptop between classes with RedFlag enabled. She visits a prepared page that resembles a social-media feed and encounters a post claiming that the university has canceled all classes the next day. Before forwarding the post to her classmates, she continues reading while RedFlag extracts the supported visible text and submits the claim to its analysis service.

RedFlag compares the claim with the evidence provided for the test scenario and places a small red flag beside the questionable statement. Maya selects the flag to open the result panel, where she sees the claim under review, its credibility score, and a brief explanation that the post relies on an outdated announcement. She follows the source link to the dated university notice, compares its date with the current post, and decides not to share the claim. If the available evidence were insufficient, the result would communicate that uncertainty rather than present the claim as conclusively false.

Later in the same browsing session, Maya opens a prepared political-news example and selects its BiasStar result to review the alternative perspectives included in the prototype. She can open the supporting sources, dismiss the panel, and continue browsing without leaving a warning over the page controls. At the end of the session, Maya uses the feedback option to report any unclear explanation and completes the case-study questionnaire. The team would use her responses to assess whether she noticed the indicator, understood the evidence, and could return to browsing without disruption.

In the future, RedFlag could serve a broader group of internet users, including high school and college students, educators, journalists, older adults, and organizations that teach media literacy. The real-world product could also support additional browsers, more languages, mobile devices, and users who want optional accessibility or account-based settings. Table 1 compares the limited prototype used for the case study with the intended real-world product.

Table 1

Comparison Between RedFlag Prototype and Real-World Product

Feature	Prototype	Real-World Product
Supported platform	Google Chrome on desktop	Chrome, Edge, Firefox, and desktop application
Content detection	Prepared pages and selected supported sites	Continuous detection across supported websites and social platforms
Credibility analysis	Limited AI analysis with a controlled set of evidence sources	Scalable analysis using broader verified evidence and production monitoring
BiasStar	Basic display of selected alternative perspectives	Broader perspective discovery with refined relevance and balance controls
User settings	Local settings with no required account	Optional accounts, synchronized preferences, and accessibility settings
Data storage	Small PostgreSQL data set for sources, cache, and feedback	Encrypted cloud database with backups, retention controls, and auditing

Feature	Prototype	Real-World Product
System capacity	Small case-study group on a development server	Autoscaling services for a large and geographically distributed user base

The prototype deliberately limits platform coverage, data volume, and system capacity so the development team can test RedFlag's main interaction cycle: detecting a claim, displaying a warning, presenting evidence, and collecting user feedback. The case study will show whether those core functions are understandable and useful before the team attempts the broader capabilities of the real-world product.

4 Glossary

AI Assistant: A software agent that uses artificial intelligence to provide interactive assistance and explanations to users.

Backend Server: A system that processes data, manages application logic, and supports front-end operations.

Browser Extension: A software module that adds specific functionality to a web browser.

Misinformation: False or misleading information that is spread regardless of intent to deceive.

Natural Language Processing (NLP): A field of artificial intelligence that focuses on the interaction between computers and human language.

Real-Time Analysis: The ability to process and evaluate data instantly as it is received.

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